

SYNOPSIS

HireSense: AI-Powered Mock Interview Platform

Submitted in partial fulfillment of the requirements for the course

Specialization Project

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1. Abstract

HireSense is an AI-powered mock interview platform designed to help candidates improve their interview performance through intelligent, data-driven assessments. Traditional interview preparation methods lack real-time feedback and personalised evaluation, leaving candidates underprepared for real-world interviews. The proposed platform integrates Computer Vision, speech analysis, and large language model-based evaluation to assess candidates across multiple dimensions including communication clarity, technical knowledge, eye contact, facial expressions, and speaking fluency. Each interview session is analysed in real time, and the system generates a comprehensive performance report with scores, strengths, weaknesses, and personalised improvement recommendations. HireSense is built using Python, OpenCV, MediaPipe, speech recognition libraries, and an AI evaluation engine, backed by a cloud-based database. The platform aims to democratise high-quality interview coaching by making personalised assessments accessible to every candidate.

2. Problem Statement

Most job seekers lack access to personalised, data-driven interview preparation tools. Conventional methods such as peer mock interviews, self-practice, or generic online resources do not provide structured, objective feedback on verbal communication, non-verbal behaviour, and technical accuracy. This gap in preparation resources results in higher interview anxiety and lower success rates for candidates. There is a clear need for an intelligent, automated interview assessment platform that evaluates candidates holistically and guides them towards continuous improvement.

3. Objectives

1. To develop an AI-powered mock interview platform that simulates real interview environments.
2. To implement Computer Vision techniques for analysing eye contact, facial expressions, and body language.
3. To integrate speech analysis for evaluating fluency, pace, filler words, and clarity.
4. To assess technical responses using natural language processing and AI-driven evaluation.
5. To generate detailed performance reports with personalised improvement recommendations.

6. To provide candidates with a confidence-building, iterative interview preparation experience.

4. Existing System

Several platforms currently offer interview preparation support:

- Video recording tools that allow candidates to watch playback of their interviews.
- Online coding platforms with automated test evaluation (e.g., HackerRank, LeetCode).
- AI chatbots that simulate Q&A-based interview practice.
- HR software with basic video interview recording and sharing features.
- Human-led mock interview services offered by coaching institutes.

While these tools offer partial support, none of them provide an integrated, multi-modal assessment combining visual, verbal, and technical evaluation in a single platform.

5. Limitations of Existing Systems

- Video playback tools provide no automated feedback or scoring.
- Coding platforms evaluate only technical correctness, ignoring communication skills.
- AI chatbots lack Computer Vision capabilities and cannot analyse non-verbal behaviour.
- Human-led coaching is expensive and not scalable for large numbers of candidates
- Existing tools do not offer integrated, holistic, multi-modal performance analysis.

6. Proposed Solution

HireSense proposes a comprehensive AI-powered mock interview platform that conducts personalised interview sessions and evaluates candidate performance across multiple dimensions simultaneously.

The system will:

- Present domain-specific interview questions dynamically using an AI question engine.
- Monitor and analyse eye contact, head pose, and facial expressions via Computer Vision.

- Transcribe spoken responses and evaluate fluency, filler word usage, pace, and clarity..
- Assess technical and behavioural responses using NLP-based AI evaluation models.
- Aggregate scores across all dimensions and generate a structured performance report.
- Provide personalised feedback and actionable improvement recommendations after each session.

7. System Functionalities

1. User Registration and Profile Management Module
2. Dynamic Interview Question Engine
3. Computer Vision Analysis Module (Eye Contact, Expressions, Posture)
4. Speech Analysis and Transcription Module
5. AI-Based Response Evaluation Module
6. Performance Scoring and Report Generation Module
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8. Candidate Dashboard and Progress Tracking Module

8. Tools and Technologies

Technology	Purpose
Python	Application Development
OpenCV & MediaPipe	Computer Vision and Facial Landmark Detection
SpeechRecognition / Whisper	Audio Transcription and Speech Analysis
Anthropic Claude API	AI-Driven Response Evaluation and Feedback Generation
NLP Libraries (NLTK / spaCy)	Text Processing and Fluency Analysis
React.js	Frontend User Interface Design
Flask / FastAPI	Backend Framework and REST API Development
PostgreSQL / Firebase	Database Management and Cloud Storage
Docker	Containerisation and Deployment
GitHub	Version Control and Collaboration

9. Expected Outcomes

- A fully functional AI-powered mock interview platform accessible via web browser.
- Real-time multi-modal analysis covering verbal, non-verbal, and technical dimensions.
- Objective and consistent scoring across all evaluation parameters.
- Detailed post-interview performance reports with actionable recommendations.
- Measurable improvement in candidate confidence and interview readiness over multiple sessions.
- Scalable and accessible interview preparation for candidates across all domains.

10. Project Timeline

Week	Activity
1–2	Requirement Gathering, Domain Research, and Literature Review
3–4	System Architecture Design, Database Schema Design, and UI Wireframes
5–6	Development of Computer Vision Module (Eye Contact, Expressions)
7–8	Development of Speech Analysis and Transcription Module
9–10	Integration of AI Evaluation Engine and Feedback Generator
11	Frontend Development, API Integration, and System Testing
12	Final Testing, Bug Fixing, Documentation, and Project Review

11. Conclusion

The proposed HireSense platform aims to transform the way candidates prepare for job interviews by providing an intelligent, personalised, and multi-modal assessment experience. By combining Computer Vision, speech analysis, and AI-driven evaluation in a unified platform, HireSense addresses the significant gap in accessible, objective interview coaching tools. The project is expected to deliver a scalable solution that empowers candidates to build confidence, identify areas for improvement, and perform at their best in real-world interviews.

12. References

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